LEARNMATE – AI STUDYBUDDY

5. Project Development Phase

Implementing the modular backend, database connectivity, authentication and Gemini AI integration.

Date	21-09-26
Team ID	SWTID-2026-5714
Project Name	LearnMate - AI StudyBuddy
Maximum marks	4 Marks



Epic 3: Backend Development

The AI StudyBuddy backend is developed using **Node.js**, **Express.js**, and **MongoDB**, following a modular architecture. The application separates routing, controllers, middleware, models, and utility modules to improve code maintainability and scalability. The backend exposes secure REST APIs for authentication, study material management, AI-powered content generation, and administrative operations.

Backend Structure

Core System Script Implementations

1. **Application Entry Point (index.js)**

The **index.js** file serves as the main entry point of the application. It initializes the Express server, loads environment variables, configures middleware, establishes the

AI-powered learning assistance and smart study support platform

Backend Development

Key Points

- Node.js and Express.js provide the RESTful backend.
- The index.js entry point initializes Express, loads environment variables, configures middleware, connects MongoDB and registers routes.



Reference visual from the supplied LearnMate project document.

- The database utility establishes MongoDB connectivity using Mongoose.
- Authentication middleware verifies JWT tokens and applies role-based authorization.

AI & Upload Development

Key Points

- The Gemini utility integrates Google Gemini 2.5 Flash and generates summaries, flashcards, quizzes and study plans.
- Upload middleware handles study material uploads and validates incoming files.
- AI services are kept in a dedicated layer to support maintainability and scalability.

Database Development

Key Points

- User schema stores name, email, password and role.
- Study Material schema stores uploaded learning content.
- MongoDB stores user information, study materials and AI-generated learning resources.